

dos and don'ts of handling EVs to better address range issues, concerns about the charging infrastructure are real. Social media is replete with reports of chargers not functioning or just not being available as shown on a map. Worse, a host of consumer complaints like vehicles stalling in the middle of traffic or facing a complete software fadeout might well be directly related to the quality of the charging infrastructure.

Even in a grid considered 'clean', hazards like surges and transients (short-duration disturbances in voltage or current) will occur, and their frequency or severity is something that cannot be modelled. Besides insulation failures in integrated circuits that can ultimately lead to total failure and even fires, transients are known to potentially cause significant software malfunctions ranging from system lock-up, memory corruption or even erroneous device resets. Despite the internationally accepted regulations for surge and transient immunity specifically codified in IEC 61000-4-5, mandated for charging infrastructure in India, there is little clarity on compliance and conformity checks. EV consumers using charging infrastructure could potentially end up taking on risks they never bargained for. This will be one major challenge the E2W industry, along with the government and charging infrastructure players, has to find solutions for as it guns for exponential growth.

For starters, charging infra players may have to be mandated to design hybrid installations that isolate even on-grid installations with in-line storage systems or provide active power filters. The other imperative would be to create an independent real-time physical quality assurance capability that would check installations periodically, but at random, and provide a health certificate. Going further, the government would be well-advised to immediately mandate only India-manufactured electronics hardware and software in EV charging infrastructure. 'Kill switches' and 'backdoors', remotely controlled from China,